HUJI Applicative Research Report — Exact & Social Sciences — Week 42, 2026

1 paper(s) · Exact & Social Sciences · Weekly · Week 42, 2026 · Generated October 12, 2026

■■ There is no high-potential applicable research this period — only lower-scoring papers were found. Presenting the two best below.

This week's digest highlights a significant advancement in material science from Hebrew University, focusing on novel superstructures for next-generation electronics. Researchers have developed a method to create highly stable and precise carbon nanotube components by leveraging specialized chiral polymer wraps. This breakthrough promises to enhance the performance and reliability of advanced sensors and components critical for quantum computing and high-performance electronic devices, addressing a critical need for superior materials in cutting-edge technologies.

RESEARCHERS & RESEARCH SUBJECTS — BEST AVAILABLE

• **Yossi Paltiel** — Materials, Quantum, Other

EXACT & SOCIAL SCIENCES — RESEARCH HIGHLIGHTS

1. Enhancing Coherence Length in Tailored Chiral Hybrid Polymer-Wrapped Single-Walled Carbon Nanotube Superstructures. 24/50

"New chiral polymer-wrapped carbon nanotubes unlock next-gen sensors and quantum computing components."

Main researcher: Yossi Paltiel · paltiel@mail.huji.ac.il
Institute of Applied Physics, The Hebrew University of Jerusalem, Jerusalem, Israel.
Published in Small (Weinheim an der Bergstrasse, Germany) · 2026-10-11

ABOUT THIS RESEARCH

Researchers have developed a method to improve the electronic coherence length of single-walled carbon nanotubes by wrapping them in specialized chiral hybrid polymers. This technique creates more stable and uniform carbon nanotube superstructures with enhanced precision in their physical properties.

COMMERCIAL ANGLE

This technology enables the production of high-performance carbon nanotube-based sensors and components for advanced electronics and quantum computing architectures.

COMMERCIALISATION METRICS

Novelty 7/10
The concept of tailoring chiral polymer wrapping to extend coherence length in SWCNT superstructures is innovative and extends prior work on polymer–nanotube interactions, but the novelty is moderate as similar strategies have been explored in related nanomaterials.

Commercial Potential 3/10
The work appears to be fundamental materials science aimed at improving coherence length in carbon nanotube superstructures; it does not directly describe a clear product, service, or licensable technology with immediate commercial application.

Market Size 5/10
The technology targets a specialized niche within nanomaterials and advanced electronics, suggesting a moderate but not massive addressable market; it is unlikely to be a mainstream consumer product yet.

Tech Readiness 2/10
The title describes fundamental materials science research focused on structural properties of carbon nanotubes, indicating basic lab-stage exploration of coherence length rather than a developed application or prototype.

IP Strength 7/10
The work proposes a novel polymer-wrapped SWCNT superstructure that could improve coherence length, suggesting a potentially patentable material and process, but without detailed methodology or evidence of industrial applicability the defensibility remains moderate.

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